

9th World Mathematics Team Championship 2018

Intermediate Level Individual Round 1

English Version

Instruction: This round has 15 questions (**20 minutes**).

Each question is worth 2 points. No point penalty for submitting wrong answer.

Blank answer will be assigned 0.5 point.

1. The number of positive integer divisors of 38808 equals:

- A) 144 B) 72 C) 36 D) 108 E) 180

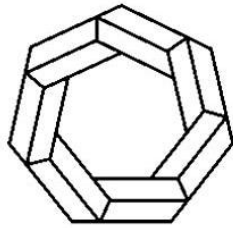
2. Find x if $2^{x+1} \cdot 4^{x+2} \cdot 8^{x+3} = 16^{x+4}$.

- A) 1 B) 6 C) 3 D) 2 E) 5

3. If $\left(1 - \frac{1}{2^2}\right) \cdot \left(1 - \frac{1}{3^2}\right) \cdots \left(1 - \frac{1}{100^2}\right) = \frac{a}{b}$ where a and b are relatively prime integers, find $a + b$.

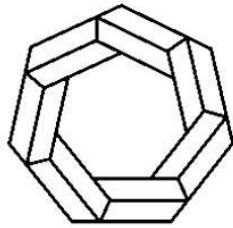
- A) 200 B) 300 C) 301 D) 302 E) 103

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4. Find $\frac{1+\sqrt{12}}{1-\sqrt{12}} + \frac{\sqrt{48}}{\sqrt{121}} + \frac{2}{11}$.
- A) 3 B) -2 C) $\sqrt{3}$ D) -1 E) $2\sqrt{3}$
5. Let O be the intersection point of the diagonals AC and BD of quadrilateral $ABCD$. If $S_{ABO} = 35 \text{ cm}^2$, $S_{BCO} = 42 \text{ cm}^2$ and $S_{CDO} = 48 \text{ cm}^2$, find S_{ADO} .
- A) 28 cm^2 B) 32 cm^2 C) 40 cm^2 D) 48 cm^2 E) 56 cm^2
6. If $(a+bx)^3 + (a-bx)^3 = -16 - 108x^2$, find $a^2 + b^2$.
- A) 5 B) 8 C) 10 D) 13 E) 25
7. Find the shortest altitude of a triangle with side lengths 13 cm, 20 cm and 21 cm.
- A) 8 cm B) 9 cm C) 10 cm D) 11 cm E) 12 cm
8. The number of 3-digit numbers divisible by 7 but not divisible by 3 equals:
- A) 87 B) 90 C) 88 D) 91 E) 85



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9. Consider all words composed by two letters a and b . Find the length of the longest word w such that none of aba , aab , abb , $aaaa$ and $bbbb$ is a sub word of w .

A) 5 B) 6 C) 7 D) 10 E) 2018

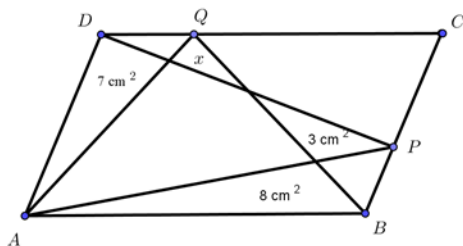
10. The least value of $2x^4 - 2x^2y^2 + y^4 - 8x^2 + 20$ equals:

A) 2 B) 6 C) 4 D) 8 E) 3

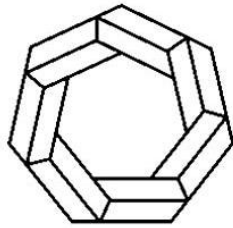
11. Find $2018^3 - 2019^3 + 1 + 3 \times 2018 \times 2019$.

A) 0 B) 2018^2 C) 2019^2 D) 3 E) 2018

12. Points P and Q lie on the sides BC and CD of a parallelogram $ABCD$. The areas of three of the regions are shown on the picture. Find x (the area of the corresponding region).

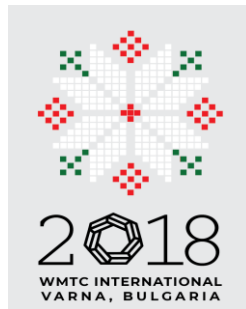
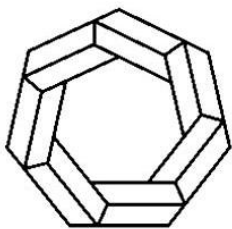


A) 1 cm^2 B) 2 cm^2 C) 3 cm^2 D) 4 cm^2 E) 5 cm^2



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13. The remainder when $3 - 3^2 + 3^3 - 3^4 + \dots - 3^{2016} + 3^{2017} - 3^{2018}$ is divided by 60 equals:
- A) 8 B) 10 C) 2 D) 54 E) 12
14. If $A = \frac{(2^1 + 3^1)(2^2 + 3^2)(2^4 + 3^4)(2^8 + 3^8)(2^{16} + 3^{16})(2^{32} + 3^{32})}{3^{60}}$, find the closest integer to A .
- A) 84 B) 80 C) 90 D) 81 E) 79
15. Find the sum of all prime divisors of $10^3 + 11^3 + \dots + 20^3$.
- A) 30 B) 33 C) 36 D) 37 E) 38



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Intermediate Level Individual Round 2

English Version

Instructions: This round has 8 questions (**40 minutes**).

Question numbers 1, 2, 3, 4, 5 and 6 are worth 4 points each.

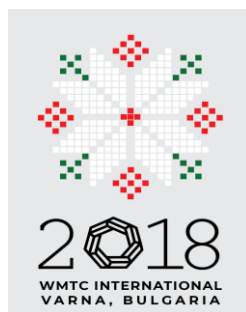
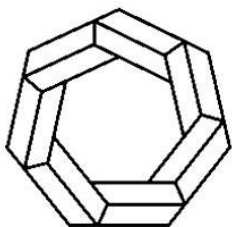
Question numbers 7 and 8 are worth 8 points each.

No point penalty for submitting wrong answers.

1. Car with 4 wheels and 1 spare tire travelled for 1000 kilometers. Denote the five tires used by A, B, C, D and E . Tires A and B have been used at the same time on the car for 550 km and tires C and D have been used at the same time on the car for 700 km. Determine how many kilometers tire E has been in use.
2. For a quadrilateral $ABCD$ we have $BC = CD = DA$ and $\angle BAD + \angle ABC = 120^\circ$. If AC and BD intersect at point O find $\angle AOB$.
3. For the sequence $a_1, a_2, \dots, a_{2018}$, where $a_1 = 2$ and $a_{n+1} = \frac{a_n^2 + 1}{2}$, find

$$a_{2018}^2 - (a_1 - 1)^2 - (a_2 - 1)^2 - \dots - (a_{2018} - 1)^2.$$

Intermediate Level Individual Round 2



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4. In a triangle ABC the angle between altitudes from A and B equals 100° . Find the angle between the angular bisectors of angle A and angle B .
5. A 3-digit number $\overline{abc}$ is called *good* if $\overline{abc} + \overline{cba}$ has only odd digits. Find the number of good integers.
6. Points P and Q on the sides AB and AC of right triangle ABC ($\angle B = 90^\circ$) are such that $AP = 2PB$ and $AQ = QC$. If $CP = 2018$ cm find the length of PQ .
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7. Amy has 5 different rings. She wants to put them on the four fingers (without the thumb) of her right hand. In how many ways can she do this? (The order of the rings on a finger matters and it is not obligatory for each finger to have a ring.)
8. To prepare for WMTC a student solved n problems during 31 day period. It is known that:
- From day 1 to day 14 inclusive he solved in average 5 problems per day
 - From day 12 to day 31 inclusive he solved in average 3 problems per day
- Find the number of different values of n for which this is possible.

Intermediate Level Relay Round 1A

R1-A Three dice are rolled. The probability that the sum of their points is 14 is written in the form $\frac{p}{q}$ where p and q are relatively prime, find $p + q$.

Intermediate Level Relay Round 1B

R1-B $T = \text{TNYWR}$ (The Number You Will Receive). In an acute triangle ABC with $\angle ACB = T^\circ$ point M is the midpoint of AB . If P and Q are points on BC and AC respectively such that $AP \perp BC$ and $BQ \perp AC$ find $\angle PMQ$.

Intermediate Level Relay Round 2A

R2-A Find the greatest value of $\left| |a - b - c| - c \right| + |a - b - c|$, where a, b, c are the numbers 2012, 2013 and 2014 in some order.

Intermediate Level Relay Round 2B

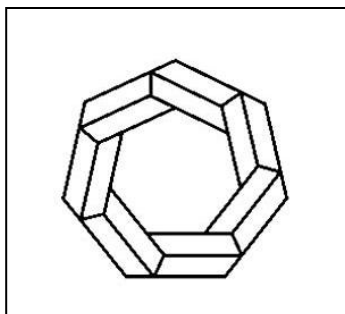
R2-B $T = \text{TNYWR}$ (The Number You Will Receive). Two points A and B stay at diametrically opposite points on a circle of center O . They start moving in opposite directions (A clockwise, B anti-clockwise). Point B moves twice faster than A . They meet for T -th time at some point C . Find the $\angle AOC$ in degrees.

Intermediate Level Relay Round 3A

R3-A Let $x = \sqrt{3 - \sqrt{4 + \sqrt{12}}} + \sqrt{3 + \sqrt{4 - \sqrt{12}}}$. Find $5x^2$.

Intermediate Level Relay Round 3B

R3-B $T = \text{TNYWR}$ (The Number You Will Receive). In a group of T students some pairs of students exchange presents. It is known that there are no three students A, B and C such that A and B exchange presents; B and C exchange presents; A and C exchange presents. Find the maximum number of presents given.



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Intermediate Level Team Round

English Version

Instruction: This round has 14 questions (**40 minutes**).

Each question is worth 5 points. No point penalty for submitting wrong answer.

1. Given five distinct positive integers such that all possible sums of two of these numbers are distinct. Find the least possible value of the largest of these five numbers.
2. For a positive integer n by $S(n)$ we denote the sum of the digits of n . Find the least value of n for which $S(n)S(n+1) = 2018$.
3. Given an isosceles triangle ABC with $\angle C = 90^\circ$ and $AC = BC = 6$. Let Y and X be points on AC and BC respectively so that $CY = CX = 2$. If D is the intersection point of AX and BY find the area of $\triangle ADB$.
4. Find the number of words of length 7 composed by two letters a and b such that there are no two adjacent letters a .
5. Some of the cells of 5×5 table are colored. What is the minimum number of cells having even number of colored adjacent cells (two cells are adjacent if they share a common side)?
6. Water is pouring from a spring at a constant speed of 40 liters per minute. Ten elephants drink all the water evenly distributed the amount of water among them. After 1 minute two elephants decided they had drunk enough and left. After one more minute three more elephants left. After the third minute the remaining 5 elephants also left. Find the amount of water each of the last 5 elephants has drunk for the entire 3 minutes.

7. Find the angle between clock arrows in 7 hr 38 min. (Using a 12-hour analog clock.)
8. Point M on the side AB and point N on the side AC of triangle ABC are such that $AN = NM = MC = CB$ and $\angle BAC = 20^\circ$. Find $\angle BNM$.
9. For what value of parameter a the equation $|2x - 1| + |3x - 2| = a - x$ has infinitely many solutions?

T = the answer of Problem #7

10. If $\overline{bcT}$ is divisible by 3, $\overline{Tbc}$ is divisible by 4 and $\overline{cTb}$ is divisible by 5 find c .

T = the answer of Problem #9

11. Find the number of pairs (x, y) of positive integers for which $x + y = Txy - 7$.

T = the answer of Problem #1

12. In a football (soccer) tournament with seven teams every two teams played one game against each other. The total number of points is divisible by $2T$. Find the number of wins in this tournament.

(In football for win the team gets 3 points, for draw gets 1 point and for loss gets 0 points)

T = answer of Problem #3

13. Find the largest positive integer k for which 3^k is a divisor of

$$3 \times 33 \times 333 \times \dots \times \underbrace{33\dots33}_T.$$

S = the answer of Problem #6; T = the answer of Problem #8

14. A student runs along a tram route with constant speed. The trams run in the two directions with constant speed through the same intervals. Every S minutes he is overtaken by a tram and every T minutes he meets a tram. Find the ratio of the speed of the student and the speed of the tram.